Worksheet 2.4 — Adding & Subtracting Rational Expressions

Grade 11 Functions, University (MCR3U) · Unit 2: Equivalent Algebraic Expressions

The Big Ideas

Fractions can only be added when they count equal-sized pieces, so we rewrite them over a **lowest common denominator (LCD)** found by *factoring* the denominators. After combining, factor the numerator, cancel, and keep the restrictions from *every* original denominator.

- **LCD:** factor every denominator; use each different factor at its *highest power*, counting shared factors once.
- **Build equivalent fractions:** multiply top and bottom by the factors a fraction is missing.
- **Subtracting:** bracket the whole second numerator — the minus sign changes *every* term inside.
- **Opposites:** $\dfrac{a}{x-4}$ and $\dfrac{b}{4-x}$ have a common denominator once you write $4-x=-(x-4)$.
- **Work rates and parallel circuits:** rates and reciprocals add: $\dfrac{1}{x}+\dfrac{1}{y}=\dfrac{x+y}{xy}$.

Concept Essentials

The lowest common denominator

Factor every denominator. The LCD contains each different factor at its **highest power**; a factor shared by two denominators is counted *once*. Multiply top and bottom of each fraction by what it is missing.

Subtracting safely

Put the whole second numerator in brackets: $\dfrac{a}{D}-\dfrac{b+c}{D}=\dfrac{a-(b+c)}{D}$. After combining, **factor the numerator** and cancel; keep the restrictions of every original denominator.

Opposites and compound fractions

$4-x=-(x-4)$ turns opposite denominators into equal ones. For a fraction inside a fraction, combine top and bottom separately, then multiply by the reciprocal.

Worked Examples

Example 1 — Same denominator

Simplify $\dfrac{9x+4}{2x-1}-\dfrac{3x-5}{2x-1}$.

Solution. $\dfrac{(9x+4)-(3x-5)}{2x-1}=\dfrac{9x+4-3x+5}{2x-1}$.

$=\dfrac{6x+9}{(2x-1)}=\dfrac{3(2x+3)}{2x-1}$, $x\neq\frac{1}{2}$.

Example 2 — Monomial denominators

Simplify $\dfrac{7}{10x}+\dfrac{3}{4x^3}$.

Solution. $10x=2\cdot 5\cdot x$ and $4x^3=2^2\cdot x^3$, so LCD $=2^2\cdot 5\cdot x^3=20x^3$.

$\dfrac{7\cdot 2x^2}{20x^3}+\dfrac{3\cdot 5}{20x^3}=\dfrac{14x^2+15}{20x^3}$, $x\neq 0$.

Example 3 — Two binomial denominators

Simplify $\dfrac{5}{x+1}+\dfrac{2}{x-4}$.

Solution. LCD $(x+1)(x-4)$.

$\dfrac{5(x-4)+2(x+1)}{(x+1)(x-4)}=\dfrac{5x-20+2x+2}{(x+1)(x-4)}=\dfrac{7x-18}{(x+1)(x-4)}$, $x\neq -1,4$.

Example 4 — Subtracting

Simplify $\dfrac{3}{x-6}-\dfrac{2}{x+1}$.

Solution. $\dfrac{3(x+1)-2(x-6)}{(x-6)(x+1)}$. Bracket the second product!

$3x+3-2x+12=x+15$: $\dfrac{x+15}{(x-6)(x+1)}$, $x\neq 6,-1$.

Example 5 — Factor the denominators first

Simplify $\dfrac{4}{x^2-4}+\dfrac{1}{x^2-x-2}$.

Solution. $x^2-4=(x-2)(x+2)$; $x^2-x-2=(x-2)(x+1)$. LCD $(x-2)(x+2)(x+1)$ — $(x-2)$ is counted once.

Numerator: $4(x+1)+1(x+2)=5x+6$.

$\dfrac{5x+6}{(x-2)(x+2)(x+1)}$, $x\neq 2,-2,-1$.

Example 6 — Opposite denominators

Simplify $\dfrac{7}{x-5}-\dfrac{2}{5-x}$.

Solution. $\dfrac{2}{5-x}=-\dfrac{2}{x-5}$, so $\dfrac{7}{x-5}+\dfrac{2}{x-5}=\dfrac{9}{x-5}$, $x\neq 5$.

Example 7 — Combine, then simplify

Simplify $\dfrac{x}{x-5}-\dfrac{25}{x^2-5x}$.

Solution. $x^2-5x=x(x-5)$; LCD $x(x-5)$.

$\dfrac{x\cdot x-25}{x(x-5)}=\dfrac{(x-5)(x+5)}{x(x-5)}=\dfrac{x+5}{x}$, $x\neq 0,5$.

Example 8 — Three terms

Simplify $\dfrac{2}{x}+\dfrac{3}{x+2}-\dfrac{4}{x(x+2)}$.

Solution. LCD $x(x+2)$: $\dfrac{2(x+2)+3x-4}{x(x+2)}=\dfrac{5x}{x(x+2)}=\dfrac{5}{x+2}$, $x\neq 0,-2$.

Example 9 — A compound fraction

Simplify $\dfrac{x-\frac{4}{x}}{1-\frac{2}{x}}$.

Solution. Top: $\dfrac{x^2-4}{x}$. Bottom: $\dfrac{x-2}{x}$. Divide: $\dfrac{x^2-4}{x}\times\dfrac{x}{x-2}=\dfrac{(x-2)(x+2)}{x-2}=x+2$, with $x\neq 0,2$.

Example 10 — Application: working together

Alex mows a lawn in x hours and Bea takes $(x+4)$ hours. Write the time together, and find it when Alex takes 6 hours.

Solution. Rates add: $\dfrac{1}{x}+\dfrac{1}{x+4}=\dfrac{2x+4}{x(x+4)}$. Time $=\dfrac{x(x+4)}{2x+4}$.

For $x=6$: $\dfrac{60}{16}=3.75$ h. Check: $\frac{1}{6}+\frac{1}{10}=\frac{16}{60}$ and $\frac{60}{16}=3.75$ ✓.

Example 11 — Test the shortcut

Is $\dfrac{1}{x}+\dfrac{1}{3}=\dfrac{2}{x+3}$? Compare with $\dfrac{x+3}{3x}$.

x	1	2	3	6
$\frac{1}{x}+\frac{1}{3}$	4/3	5/6	2/3	1/2
$\frac{2}{x+3}$	1/2	2/5	1/3	2/9
$\frac{x+3}{3x}$	4/3	5/6	2/3	1/2
same?	×	×	×	×

Solution. The second row disagrees with the first at every value tested, so $\frac{2}{x+3}$ is *not* equivalent: you never add denominators. The third row, built over the LCD $3x$, matches the first in every column.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Simplify $\frac{8x+1}{3x-2} - \frac{2x+7}{3x-2}$ and state the restriction.

Question 2

Simplify $\frac{5}{8x^2} + \frac{3}{4x}$.

Question 3

Simplify $\frac{6}{x+2} + \frac{1}{x-3}$ and state the restrictions.

Question 4

Simplify $\frac{2}{x-1} - \frac{5}{x+4}$ and state the restrictions.

Question 5

Simplify $\dfrac{3}{x^2-16}+\dfrac{2}{x^2+x-12}$. State the LCD first.

Question 6

Simplify $\dfrac{x-1}{x-6}+\dfrac{x+8}{6-x}$.

Question 7

Simplify $\dfrac{x}{x+3}-\dfrac{9}{x^2+3x}$. Factor the final numerator.

Question 8

Show that $\dfrac{1}{x-2}-\dfrac{1}{x+2}-\dfrac{4}{x^2-4}$ equals 0 for every allowed x. State the restrictions.

Question 9

Simplify the compound fraction $\dfrac{1+\frac{3}{x}}{1-\frac{9}{x^2}}$ and state the restrictions.

Question 10

Pipe A fills a pool in x hours and pipe B in $(x+6)$ hours. Write the time with both open as one fraction and evaluate it when $x = 6$.

Question 11

A student writes $\dfrac{2}{x}+\dfrac{3}{4}=\dfrac{5}{x+4}$. Test the claim at $x = 4$, explain the error, and give the correct sum.

Question 12

Find the LCD of $x^2 - x$ and $x^2 - 1$, then simplify $\dfrac{1}{x^2 - x} + \dfrac{1}{x^2 - 1}$.

Question 13

Two resistors x and $3x$ ohms are in parallel: $\dfrac{1}{R} = \dfrac{1}{x} + \dfrac{1}{3x}$. Find R in terms of x, and its value for $x = 8$.

Question 14 — Challenge

Show that $\dfrac{1}{n} - \dfrac{1}{n+1} = \dfrac{1}{n(n+1)}$. Use it to evaluate $\dfrac{1}{2 \cdot 3} + \dfrac{1}{3 \cdot 4} + \cdots + \dfrac{1}{19 \cdot 20}$.

Question Recap & Answer Key

Question Recap

1. Simplify $\dfrac{8x+1}{3x-2}-\dfrac{2x+7}{3x-2}$ and state the restriction.
2. Simplify $\dfrac{5}{8x^2}+\dfrac{3}{4x}$.
3. Simplify $\dfrac{6}{x+2}+\dfrac{1}{x-3}$ and state the restrictions.
4. Simplify $\dfrac{2}{x-1}-\dfrac{5}{x+4}$ and state the restrictions.
5. Simplify $\dfrac{3}{x^2-16}+\dfrac{2}{x^2+x-12}$. State the LCD first.
6. Simplify $\dfrac{x-1}{x-6}+\dfrac{x+8}{6-x}$.
7. Simplify $\dfrac{x}{x+3}-\dfrac{9}{x^2+3x}$. Factor the final numerator.
8. Show that $\dfrac{1}{x-2}-\dfrac{1}{x+2}-\dfrac{4}{x^2-4}$ equals 0 for every allowed x. State the restrictions.
9. Simplify the compound fraction $\dfrac{1+\frac{3}{x}}{1-\frac{9}{x^2}}$ and state the restrictions.
10. Pipe A fills a pool in x hours and pipe B in $(x+6)$ hours. Write the time with both open as one fraction and evaluate it when $x=6$.
11. A student writes $\dfrac{2}{x}+\dfrac{3}{4}=\dfrac{5}{x+4}$. Test the claim at $x=4$, explain the error, and give the correct sum.
12. Find the LCD of x^2-x and x^2-1, then simplify $\dfrac{1}{x^2-x}+\dfrac{1}{x^2-1}$.
13. Two resistors x and $3x$ ohms are in parallel: $\dfrac{1}{R}=\dfrac{1}{x}+\dfrac{1}{3x}$. Find R in terms of x, and its value for $x=8$.
14. Show that $\dfrac{1}{n}-\dfrac{1}{n+1}=\dfrac{1}{n(n+1)}$. Use it to evaluate $\dfrac{1}{2\cdot3}+\dfrac{1}{3\cdot4}+\cdots+\dfrac{1}{19\cdot20}$.

✓ Answer Key

1. $\dfrac{(8x+1)-(2x+7)}{3x-2}=\dfrac{6x-6}{(3x-2)}=\dfrac{6(x-1)}{3x-2}$, $x\neq\frac{2}{3}$.
2. LCD $8x^2$: $\dfrac{5+3\cdot 2x}{8x^2}=\dfrac{6x+5}{8x^2}$, $x\neq0$.
3. $\dfrac{6(x-3)+(x+2)}{(x+2)(x-3)}=\dfrac{7x-16}{(x+2)(x-3)}$, $x\neq-2,3$.
4. $\dfrac{2(x+4)-5(x-1)}{(x-1)(x+4)}=\dfrac{-3x+13}{(x-1)(x+4)}$, $x\neq1,-4$.
5. LCD $(x-4)(x+4)(x-3)$. Numerator $3(x-3)+2(x-4)=5x-17$. $\dfrac{5x-17}{(x-4)(x+4)(x-3)}$, $x\neq4,-4,3$.
6. $\dfrac{x-1}{x-6}-\dfrac{x+8}{x-6}=\dfrac{-9}{x-6}$, $x\neq6$.
7. LCD $x(x+3)$: $\dfrac{x^2-9}{x(x+3)}=\dfrac{(x-3)(x+3)}{x(x+3)}=\dfrac{x-3}{x}$, $x\neq0,-3$.
8. LCD $(x-2)(x+2)$: $\dfrac{(x+2)-(x-2)-4}{(x-2)(x+2)}=\dfrac{0}{(x-2)(x+2)}=0$, $x\neq\pm2$.
9. $\dfrac{x+3}{x}\times\dfrac{x^2}{(x-3)(x+3)}=\dfrac{x}{x-3}$, $x\neq0,3,-3$.
10. $\dfrac{1}{x}+\dfrac{1}{x+6}=\dfrac{2x+6}{x(x+6)}$, so time $=\dfrac{x(x+6)}{2x+6}$. At $x=6$: $\dfrac{72}{18}=4$ hours.
11. At $x=4$: $\frac{2}{4}+\frac{3}{4}=\frac{5}{4}$ but $\frac{5}{8}$. Denominators are never added; use the LCD $4x$: $\dfrac{2}{x}+\dfrac{3}{4}=\dfrac{8+3x}{4x}$.
12. $x(x-1)$ and $(x-1)(x+1)$: LCD $x(x-1)(x+1)$. $\dfrac{(x+1)+x}{x(x-1)(x+1)}=\dfrac{2x+1}{x(x-1)(x+1)}$, $x\neq0,1,-1$.
13. $\dfrac{1}{R}=\dfrac{3+1}{3x}=\dfrac{4}{3x}$, so $R=\dfrac{3x}{4}$. For $x=8$: 6 ohms.
14. $\dfrac{1}{n}-\dfrac{1}{n+1}=\dfrac{(n+1)-n}{n(n+1)}=\dfrac{1}{n(n+1)}$. The sum is $(\frac{1}{2}-\frac{1}{3})+(\frac{1}{3}-\frac{1}{4})+\cdots+(\frac{1}{19}-\frac{1}{20})=\frac{1}{2}-\frac{1}{20}=\frac{9}{20}$.